US Patent & Trademark Office
Patent Public Search | Text View

United States Patent	12417203
Kind Code	B1
Date of Patent	September 16, 2025
Inventor(s)	Sonera; Yoel H. et al.

High speed, extensible and self-healing data network for AESA RFIC arrays

Abstract

An AESA may include a point-to-point serial interface between RFIC that implements positional addressing and is extensible to an unlimited amount of RFIC. A controller of the AESA may access registers of one or more RFIC and sends control and synchronization signals using a low-latency serial interface. The AESA may include primary and secondary interfaces which form a mesh network allowing communications around defective devices or interconnects and reduced area overhead resulting in array size reduction. The AESA may include health monitoring to determine location of data network faults providing improved array mean-time between failure (MTBF) and fault recovery. The AESA may allow faster antenna array configuration and beam steering rates for improved uptime and optimized scan modes. The AESA may experience improvements in manufacturability and reliability due to redundancy and fault isolation. The AESA may include more RFIC per serial interface allows scaling into larger arrays.

Inventors: **Sonera; Yoel H. (Palm Bay, FL), West; James B. (Cedar Rapids, IA), Teague; Jacob G. (West Melbourne, FL), Meinecke; James P. (Fayetteville, AR)**

Applicant: **Rockwell Collins, Inc.** (Cedar Rapids, IA)

Family ID: **1000007918790**

Assignee: **Rockwell Collins, Inc. (Cedar Rapids, IA)**

Appl. No.: **18/667476**

Filed: **May 17, 2024**

Publication Classification

Int. Cl.: **G06F13/42** (20060101); **H01Q3/26** (20060101); **H01Q3/28** (20060101); **H01Q3/36** (20060101); **H04B1/40** (20150101)

U.S. Cl.:

CPC **G06F13/4265** (20130101); **G06F13/4282** (20130101); **H01Q3/2682** (20130101); **H01Q3/28** (20130101); **H01Q3/36** (20130101); **H04B1/40** (20130101);

Field of Classification Search

CPC: G06F (13/4282); G06F (13/4265); H01Q (3/2682); H01Q (3/28); H01Q (3/36); H04B (1/40)

References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
8902101	12/2013	Sishtla et al.	N/A	N/A
9478858	12/2015	West	N/A	G01S 13/953
9831906	12/2016	Xie et al.	N/A	N/A
10833408	12/2019	Paulsen et al.	N/A	N/A
10910709	12/2020	Hill et al.	N/A	N/A
11018425	12/2020	Hageman et al.	N/A	N/A
11093429	12/2020	Munetoh	N/A	N/A
11469918	12/2021	Cui et al.	N/A	N/A
11835648	12/2022	Sishtla et al.	N/A	N/A
11886366	12/2023	Mishra et al.	N/A	N/A
2017/0118125	12/2016	Mishra	N/A	G06F 13/4282
2021/0081348	12/2020	Rodd	N/A	G06F 13/4291
2022/0308165	12/2021	Teague et al.	N/A	N/A
2023/0223707	12/2022	Franson	343/893	H01Q 21/065
2024/0347907	12/2023	Cohen	N/A	H01Q 1/2208

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
3661075	12/2019	EP	N/A
2592259	12/2020	GB	N/A

Primary Examiner: Dang; Phong H

Attorney, Agent or Firm: Suiter Swantz IP

Background/Summary

TECHNICAL FIELD

(1) The present disclosure generally relates to transmission of digital information, and more specifically to mesh networks.

BACKGROUND

(2) Active electronically scanned array (AESA) antennas used in multi-mode radars require a large amount of radio frequency integrated circuits (RFIC) interconnected through a data communications network. Most RFIC use multi-drop serial peripheral interface (SPI) busses which

are limited to a few devices per bus and slow clock rates. The multi-drop method creates signal integrity issues, single points of failure and limits the array size and control rate. The AESA antennas may be limited to four of the RFIC per multi-drop bus due to signal integrity and addressing limitations.

(3) The AESA may include tens, hundreds, or thousands of the RFIC. The AESA may then require large and complex interface bus routing between the controller and the RFIC to control the RFIC, including additional control and address signals. The number of RFIC in the AESA may be limited by the number of multi-drop busses required to control the RFIC due to space limitations in routing the multi-drop busses while maintaining spacing between adjacent RFIC. Reliability may also be reduced due to the complexity of the RFIC interface to the controller.

(4) The AESA may also experience clock speed limits due to the multi-drop SPI bus. The clock speed limits may be due to signal integrity issues from long traces, board-to-board interfaces.

(5) Therefore, it would be advantageous to provide a device, system, and method that cures the shortcomings described above.

SUMMARY

(6) In some aspects, the techniques described herein relate to an active electronically scanned array including: a controller; a plurality of radio frequency integrated circuits, wherein the plurality of radio frequency integrated circuits are arranged in a lattice; a plurality of primary point-to-point serial interfaces; and a plurality of secondary point-to-point serial interfaces, wherein each of the plurality of radio frequency integrated circuits include one primary input from the plurality of primary point-to-point serial interfaces and one secondary input from the plurality of secondary point-to-point serial interfaces, wherein each of the plurality of radio frequency integrated circuits include one primary output to the plurality of primary point-to-point serial interfaces and one secondary output to the plurality of secondary point-to-point serial interfaces; wherein the controller and the plurality of radio frequency integrated circuits are configured to transmit a plurality of network messages between the controller and the plurality of radio frequency integrated circuits over the plurality of primary point-to-point serial interfaces and the plurality of secondary point-to-point serial interfaces; wherein the plurality of network messages include: a read network message including a read command frame and a first plurality of data frames; a write network message including a write command frame and a second plurality of data frames; a write-all network message including a write-all command frame and a third plurality of data frames; and a sync network message including a sync command frame.

(7) In some aspects, the techniques described herein relate to an active electronically scanned array, wherein the read network message and the write network message use positional addressing to define an address of the first plurality of data frames and the second plurality of data frames.

(8) In some aspects, the techniques described herein relate to an active electronically scanned array, wherein the read command frame, the first plurality of data frames, the write command frame, the second plurality of data frames, the write-all command frame, and the third plurality of data frames each include a same frame size.

(9) In some aspects, the techniques described herein relate to an active electronically scanned array, wherein the first plurality of data frames immediately follow the read command frame, the second plurality of data frames immediately follow the write command frame, and the third plurality of data frames immediately follow the write-all command frame.

(10) In some aspects, the techniques described herein relate to an active electronically scanned array, wherein the plurality of radio frequency integrated circuits are configured to append the first plurality of data frames to the read network message in response to receiving the read command frame, wherein the plurality of radio frequency integrated circuits are configured to truncate the second plurality of data frames from the write network message in response to receiving the write command frame.

(11) In some aspects, the techniques described herein relate to an active electronically scanned

array, wherein the read command frame, the first plurality of data frames, the write command frame, the second plurality of data frames, the write-all command frame, and the third plurality of data frames each include a start bit, a header, and a packet payload.
(12) In some aspects, the techniques described herein relate to an active electronically scanned array, wherein the packet payload of the read command frame, the write command frame, and the write-all command frame each include a plurality of address bits for the first plurality of data frames, the second plurality of data frames, and the third plurality of data frames, respectively.
(13) In some aspects, the techniques described herein relate to an active electronically scanned array, wherein the plurality of radio frequency integrated circuits each include: a first receiver, a first decoder, a second receiver, a second decoder, a router, a plurality of device registers, a first encoder, a first transmitter, a second encoder, and a second transmitter; wherein the plurality of address bits contain a register address of one or more of the plurality of device registers.
(14) In some aspects, the techniques described herein relate to an active electronically scanned array, wherein the plurality of address bits do not indicate to and from which of the plurality of radio frequency integrated circuits the plurality of network messages are addressed.
(15) In some aspects, the techniques described herein relate to an active electronically scanned array, wherein the controller is configured to read a health status message from the radio frequency integrated circuits by sending the read network message with the read command frame including the packet payload with the address at which a health status message is stored in the plurality of device registers.
(16) In some aspects, the techniques described herein relate to an active electronically scanned array, wherein the plurality of radio frequency integrated circuits are each configured to write the packet payload of the third plurality of data frames into the plurality of device registers.
(17) In some aspects, the techniques described herein relate to an active electronically scanned array, wherein the first receiver and the second receiver receive serial input from the plurality of primary point-to-point serial interfaces and the plurality of secondary point-to-point serial interfaces, respectively.
(18) In some aspects, the techniques described herein relate to an active electronically scanned array, wherein the first decoder is coupled to the first receiver, wherein the second decoder is coupled to the second receiver, wherein the first decoder and the second decoder are each configured to decode a data signal to determine a command frame and one or more data frames of a network message.
(19) In some aspects, the techniques described herein relate to an active electronically scanned array, wherein the router is coupled to the first decoder, the second decoder, the plurality of device registers, the first decoder, and the second decoder.
(20) In some aspects, the techniques described herein relate to an active electronically scanned array, wherein the header includes a plurality of bits indicating whether a frame is the read command frame, the write command frame, the write-all command frame, the sync command frame, or a data frame.
(21) In some aspects, the techniques described herein relate to an active electronically scanned array, wherein the plurality of primary point-to-point serial interfaces and the plurality of secondary point-to-point serial interfaces include a clock signal and a data signal, wherein one of the clock signal and the data signal are on separate pins or the clock signal is encoded onto the data signal using Manchester encoding.
(22) In some aspects, the techniques described herein relate to an active electronically scanned array, wherein the plurality of primary point-to-point serial interfaces are arranged orthogonal to plurality of secondary point-to-point serial interfaces.
(23) In some aspects, the techniques described herein relate to an active electronically scanned array, including a plurality of multi-chip modules; wherein the plurality of multi-chip modules include the plurality of radio frequency integrated circuits and a plurality of transmit/receive

modules.
(24) In some aspects, the techniques described herein relate to an active electronically scanned array, wherein the plurality of radio frequency integrated circuits are arranged in a plurality of arrays, wherein the plurality of primary point-to-point serial interfaces and the plurality of secondary point-to-point serial interfaces connect the plurality of arrays.
(25) In some aspects, the techniques described herein relate to an AESA including: a controller; a plurality of multi-chip modules, wherein the plurality of multi-chip modules include: a plurality of radio frequency integrated circuits, wherein the plurality of radio frequency integrated circuits are arranged in a lattice; and a plurality of transmit/receive modules; a plurality of primary point-to-point serial interfaces; a plurality of secondary point-to-point serial interfaces, wherein each of the plurality of radio frequency integrated circuits include one primary input from the plurality of primary point-to-point serial interfaces and one secondary input from the plurality of secondary point-to-point serial interfaces, wherein each of the plurality of radio frequency integrated circuits include one primary output to the plurality of primary point-to-point serial interfaces and one secondary output to the plurality of secondary point-to-point serial interfaces; wherein the controller and the plurality of radio frequency integrated circuits are configured to transmit a plurality of network messages between the controller and the plurality of radio frequency integrated circuits over the plurality of primary point-to-point serial interfaces and the plurality of secondary point-to-point serial interfaces; wherein the plurality of network messages include: a read network message including a read command frame and a first plurality of data frames; a write network message including a write command frame and a second plurality of data frames, wherein the read network message and the write network message use positional addressing to define an address of the first plurality of data frames and the second plurality of data frames; a write-all network message including a write-all command frame and a third plurality of data frames; and a sync network message including a sync command frame.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
(1) Implementations of the concepts disclosed herein may be better understood when consideration is given to the following detailed description thereof. Such description makes reference to the included drawings, which are not necessarily to scale, and in which some features may be exaggerated and some features may be omitted or may be represented schematically in the interest of clarity. Like reference numerals in the drawings may represent and refer to the same or similar element, feature, or function. In the drawings:
(2) FIG. **1**A depicts an architecture of an AESA, in accordance with one or more embodiments of the present disclosure.
(3) FIG. **1**B depicts a multi-chip module of the AESA with an RFIC, in accordance with one or more embodiments of the present disclosure.
(4) FIG. **1**C depicts a block diagram of the RFIC, in accordance with one or more embodiments of the present disclosure.
(5) FIG. **1**D depicts command frames and data frames of a network message of the AESA, in accordance with one or more embodiments of the present disclosure.
(6) FIG. **1**E depicts a timing diagram of a read network message of the AESA with serial data input (SDI) and serial data output (SDO), in accordance with one or more embodiments of the present disclosure.
(7) FIG. **1**F depicts a timing diagram of a write network message of the AESA with SDI and SDO, in accordance with one or more embodiments of the present disclosure.
(8) FIG. **1**G depicts a timing diagram of a write-all network message of the AESA with SDI and

SDO, in accordance with one or more embodiments of the present disclosure.
(9) FIG. **1**H depicts a timing diagram of a sync network message of the AESA with SDI and SDO, in accordance with one or more embodiments of the present disclosure.
(10) FIG. **2** depicts a flow diagram of a method, in accordance with one or more embodiments of the present disclosure.
(11) FIG. **3** depicts an architecture of the AESA, in accordance with one or more embodiments of the present disclosure.

DETAILED DESCRIPTION

(12) Before explaining one or more embodiments of the disclosure in detail, it is to be understood that the embodiments are not limited in their application to the details of construction and the arrangement of the components or steps or methodologies set forth in the following description or illustrated in the drawings. In the following detailed description of embodiments, numerous specific details are set forth in order to provide a more thorough understanding of the disclosure. However, it will be apparent to one of ordinary skill in the art having the benefit of the instant disclosure that the embodiments disclosed herein may be practiced without some of these specific details. In other instances, well-known features may not be described in detail to avoid unnecessarily complicating the instant disclosure.
(13) As used herein a letter following a reference numeral is intended to reference an embodiment of the feature or element that may be similar, but not necessarily identical, to a previously described element or feature bearing the same reference numeral (e.g., **1**, **1***a*, **1***b*). Such shorthand notations are used for purposes of convenience only and should not be construed to limit the disclosure in any way unless expressly stated to the contrary.
(14) Further, unless expressly stated to the contrary, “or” refers to an inclusive or and not to an exclusive or. For example, a condition A or B is satisfied by any one of the following: A is true (or present) and B is false (or not present), A is false (or not present) and B is true (or present), and both A and B are true (or present).
(15) In addition, use of “a” or “an” may be employed to describe elements and components of embodiments disclosed herein. This is done merely for convenience and “a” and “an” are intended to include “one” or “at least one,” and the singular also includes the plural unless it is obvious that it is meant otherwise.
(16) Finally, as used herein any reference to “one embodiment” or “some embodiments” means that a particular element, feature, structure, or characteristic described in connection with the embodiment is included in at least one embodiment disclosed herein. The appearances of the phrase “in some embodiments” in various places in the specification are not necessarily all referring to the same embodiment, and embodiments may include one or more of the features expressly described or inherently present herein, or any combination or sub-combination of two or more such features, along with any other features which may not necessarily be expressly described or inherently present in the instant disclosure.
(17) Reference will now be made in detail to the subject matter disclosed, which is illustrated in the accompanying drawings.
(18) Embodiments of the present disclosure are directed to High Speed, Extensible and Self-Healing Data Network for Active Electronically Scanned Arrays (AESA) Radio frequency integrated circuit (RFIC) Arrays. The AESA may include a point-to-point serial interface between RFIC that implements positional addressing and is extensible to an unlimited amount of RFIC. A controller of the AESA may access registers of one or more RFIC and sends control and synchronization signals using a low-latency serial interface. The AESA may include primary and secondary interfaces which form a mesh network allowing communications around defective devices or interconnects and reduced area overhead resulting in array size reduction. The AESA may include health monitoring to determine location of data network faults providing improved array mean-time between failure (MTBF) and fault recovery. The AESA may allow faster antenna

array configuration and beam steering rates for improved uptime and optimized scan modes. The AESA may experience improvements in manufacturability and reliability due to redundancy and fault isolation. The AESA may include more RFIC per serial interface allows scaling into larger arrays.

(19) U.S. Pat. No. 9,478,858B1, titled "Multi-chip module architecture"; U.S. Pat. No. 10,833,408B2, titled "Electronically scanned array"; U.S. Pat. No. 10,910,709B1, titled "Control architecture for electronically scanned array"; U.S. Pat. No. 11,018,425B1, titled "Active electronically scanned array with power amplifier drain bias tapering for optimal power added efficiency"; U.S. Pat. No. 9,831,906B1, titled "Active electronically scanned array with power amplifier drain bias tapering"; are incorporated herein by reference in the entirety.

(20) FIGS. **1**A-**1**H depicts an AESA **100**, in accordance with one or more embodiments of the present disclosure. The AESA **100** may include a controller **102**, radio frequency integrated circuits (RFIC **104**), primary point-to-point serial interfaces **106**, and secondary point-to-point serial interfaces **108**.

(21) The controller **102** may be a beam steering computer for the AESA **100**. The controller **102** may include control logic. The controller **102** may provide digital input/output (I/O) to the RFIC **104**.

(22) The AESA **100** may include any number of the RFIC **104**. For example, the AESA **100** is depicted as including sixteen of the RFIC **104**, although this is not intended to be limiting. The RFIC **104** may be arranged in a periodic array. The periodic array may be periodic along one or more rows and/or along one or more columns. The RFIC **104** may be arranged in a lattice. For example, the RFIC **104** may be arranged in a square lattice or the like. The lattice may include rows and columns of the RFIC **104**. The rows of the RFIC **104** may be along the horizontal. The columns of the RFIC **104** may be along the vertical. The rows and columns of the RFIC **104** may be spaced apart at equal distances where the lattice is the square lattice.

(23) The RFIC **104** may be directly or indirectly connected to the controller **102**. The primary point-to-point serial interfaces **106** and/or the secondary point-to-point serial interfaces **108** may connect between pairs of the controller **102** and the RFIC **104** and/or pairs of the RFIC **104**. Each of the primary point-to-point serial interfaces **106** and/or the secondary point-to-point serial interfaces **108** may only connect between two points (e.g., the controller **102** and one of the RFIC **104**; two of the RFIC **104**). The primary point-to-point serial interfaces **106** and/or the secondary point-to-point serial interfaces **108** do not connect between more than two of the points.

(24) The primary point-to-point serial interfaces **106** and/or the secondary point-to-point serial interfaces **108** may be digital interconnects.

(25) The primary point-to-point serial interfaces **106** and/or the secondary point-to-point serial interfaces **108** may be from the controller **102** to the RFIC **104**. The primary point-to-point serial interfaces **106** and/or the secondary point-to-point serial interfaces **108** from the controller **102** to the RFIC **104** may provide serial data from the controller **102** to the RFIC **104**. The primary point-to-point serial interfaces **106** and/or the secondary point-to-point serial interfaces **108** from the controller **102** to the RFIC **104** may also be referred to as Master-out, Slave-in (MOSI). The controller **102** may include one or more primary outputs (PRI) to the primary point-to-point serial interfaces **106** and one or more secondary outputs (SEC) to the secondary point-to-point serial interfaces **108**. The one or more primary outputs to the primary point-to-point serial interfaces **106** and the one or more secondary outputs to the secondary point-to-point serial interfaces **108** may be primary inputs and secondary inputs, respectively, to one or more of the RFIC **104**.

(26) The primary point-to-point serial interfaces **106** and/or the secondary point-to-point serial interfaces **108** may be from the RFIC **104** to the RFIC **104**. The primary point-to-point serial interfaces **106** and/or the secondary point-to-point serial interfaces **108** from the RFIC **104** to the RFIC **104** may provide serial data from the RFIC **104** to the RFIC **104**. The primary point-to-point serial interfaces **106** and/or the secondary point-to-point serial interfaces **108** from the RFIC **104** to

the RFIC **104** may also be referred to as Slave-out, Slave-in (SOSI).

(27) Each of the RFIC **104** may include one primary input from the primary point-to-point serial interfaces **106** and one secondary input from the secondary point-to-point serial interfaces **108**. Each of the RFIC **104** may also include one primary output to the primary point-to-point serial interfaces **106** and one secondary output to the secondary point-to-point serial interfaces **108**. The one primary input to the RFIC **104** from the primary point-to-point serial interfaces **106** may either be the primary output from the controller **102** to the primary point-to-point serial interfaces **106** or the primary output from a previous of the RFIC **104** to the primary point-to-point serial interfaces **106**. Similarly, the one secondary input to the RFIC **104** from the secondary point-to-point serial interfaces **108** may either be the secondary output from the controller **102** to the secondary point-to-point serial interfaces **108** or the secondary output from a previous of the RFIC **104** to the secondary point-to-point serial interfaces **108**.

(28) Each of the RFIC **104** may have at least two inputs and at least two outputs. For example, each of the RFIC **104** may have an input from the primary point-to-point serial interfaces **106**, an input from the secondary point-to-point serial interfaces **108**, an output to the primary point-to-point serial interfaces **106**, and an output to the secondary point-to-point serial interfaces **108**.

(29) The primary point-to-point serial interfaces **106** and/or the secondary point-to-point serial interfaces **108** may be from the RFIC **104** to the controller **102**. The primary point-to-point serial interfaces **106** and/or the secondary point-to-point serial interfaces **108** from the RFIC **104** to the controller **102** may provide serial data from the RFIC **104** to the controller **102**. The primary point-to-point serial interfaces **106** and/or the secondary point-to-point serial interfaces **108** from the RFIC **104** to the controller **102** may also be referred to as Master-in, Slave-out (MISO).

(30) The one primary output from the RFIC **104** to the primary point-to-point serial interfaces **106** may either be the primary input to the controller **102** from the primary point-to-point serial interfaces **106** or the primary input to a next of the RFIC **104** from the primary point-to-point serial interfaces **106**. Similarly, the one secondary output from the RFIC **104** to the secondary point-to-point serial interfaces **108** may either be the secondary input to the controller **102** from the secondary point-to-point serial interfaces **108** or the secondary input to a next of the RFIC **104** from the secondary point-to-point serial interfaces **108**.

(31) The controller **102** may include one or more primary inputs from the primary point-to-point serial interfaces **106** and/or one or more secondary inputs from the secondary point-to-point serial interfaces **108**. The one or more primary inputs to the controller **102** from the primary point-to-point serial interfaces **106** may be the primary output from the RFIC **104** to the primary point-to-point serial interfaces **106**. Similarly, the one or more primary inputs to the controller **102** from the secondary point-to-point serial interfaces **108** may be the secondary output from the RFIC **104** to the secondary point-to-point serial interfaces **108**.

(32) Where the primary input and the secondary input are received from previous of the RFIC **104**, none of the RFIC **104** may receive both the primary input and the secondary input from the same of the previous of the RFIC **104**. Similarly, where the primary output and the secondary output are to a next of the RFIC **104**, none of the RFIC **104** may output both the primary output and the secondary output to the same of the next of the RFIC **104**. Thus, the input of the RFIC **104** may from unique pairs of the controller **102** and/or the RFIC **104** and/or the output from the RFIC **104** may be to unique pairs of the controller **102** and/or the RFIC **104**.

(33) The primary point-to-point serial interfaces **106** and/or the secondary point-to-point serial interfaces **108** may connect between the controller **102** and the RFIC **104** in a grid topology. The grid topology may provide a redundant serial interface architecture for the inputs to and the outputs from the RFIC **104**.

(34) The AESA **100** may include rows of the primary point-to-point serial interfaces **106** and/or columns of the secondary point-to-point serial interfaces **108**. The primary point-to-point serial interfaces **106** and/or the secondary point-to-point serial interfaces **108** may be routed horizontally

and vertically, respectively, to form the grid topology. The primary point-to-point serial interfaces **106** may be arranged orthogonal to secondary point-to-point serial interfaces **108**. The primary point-to-point serial interfaces **106** may be arranged orthogonal to secondary point-to-point serial interfaces **108** such that each of the RFIC **104** may be controlled by either the primary point-to-point serial interfaces **106** and/or the secondary point-to-point serial interfaces **108**. The primary point-to-point serial interfaces **106** may be independent of the secondary point-to-point serial interfaces **108**. The primary point-to-point serial interfaces **106** may not share a common path with the secondary point-to-point serial interfaces **108**.

(35) The AESA **100** may include any number of the RFIC **104** in the columns and/or rows. The RFIC **104** may form an M-by-N array of the RFIC **104**, where M and N are integers which are two or greater. M and N may or may not be the same. For example, the AESA **100** is depicted as including M=four of the RFIC **104** along the horizontal and N=four of the RFIC **104** along the vertical, thereby forming a four-by-four array of the RFIC **104**. The AESA **100** may include at four rows of the RFIC **104** and/or at least four columns of the RFIC **104**. The RFIC **104** may be arranged in at least a four-by-four array. It is contemplated that the AESA **100** may include up to a sixteen-by-sixteen array of the RFIC **104**, or more. The number of the RFIC **104** in the rows and/or columns is only limited by the physical size of the AESA **100** and the desired update rate.

(36) Rows of the primary point-to-point serial interfaces **106** and/or columns of the secondary point-to-point serial interfaces **108** may connect between the controller **102** and the RFIC **104** across power domains, or other common failure mode elements. The primary point-to-point serial interfaces **106** and/or the secondary point-to-point serial interfaces **108** may connect between the controller **102** and the RFIC **104** across power domains, or other common failure mode elements to avoid single points of failure.

(37) The number of the primary point-to-point serial interfaces **106** may be based on the number of the RFIC **104**. In the example of the four-by-four array of the RFIC **104**, the AESA **100** may include four of the primary point-to-point serial interfaces **106** between the controller **102** and the first column of the RFIC **104**, four of the primary point-to-point serial interfaces **106** between the first column of the RFIC **104** and the second column of the RFIC **104**, four of the primary point-to-point serial interfaces **106** between the second column of the RFIC **104** and the third column of the RFIC **104**, four of the primary point-to-point serial interfaces **106** between the third column of the RFIC **104** and the fourth column of the RFIC **104**, and four of the primary point-to-point serial interfaces **106** between the fourth column of the RFIC **104** and the controller **102**, for a total of twenty of the primary point-to-point serial interfaces **106**.

(38) The number of the secondary point-to-point serial interfaces **108** may be based on the number of the RFIC **104**. In the example of the four-by-four array of the RFIC **104**, the AESA **100** may include four of the secondary point-to-point serial interfaces **108** between the controller **102** and the first row of the RFIC **104**, four of the secondary point-to-point serial interfaces **108** between the first row of the RFIC **104** and the second row of the RFIC **104**, four of the secondary point-to-point serial interfaces **108** between the second row of the RFIC **104** and the third row of the RFIC **104**, four of the secondary point-to-point serial interfaces **108** between the third row of the RFIC **104** and the fourth row of the RFIC **104**, and four of the secondary point-to-point serial interfaces **108** between the fourth row of the RFIC **104** and the controller **102**, for a total of twenty of the secondary point-to-point serial interfaces **108**.

(39) Connecting between the controller **102** and the rows of the RFIC **104** using the primary point-to-point serial interfaces **106** and/or the connecting between the controller **102** and the columns of the RFIC **104** using the secondary point-to-point serial interfaces **108** may enable extending the number of the RFIC **104** which are controlled by the controller **102**. For example, the AESA **100** may enable larger arrays of the RFIC **104** due to not being limited to four of the RFIC **104** per multi-drip SPI bus.

(40) The AESA **100** may include multi-chip modules (MCM **126**). The AESA **100** may include an

array of the MCM **126**. For example, the MCM **126** may be arranged in the rows and the columns in the lattice to define the lattice of the RFIC **104**. The MCM **126** may include the RFIC **104** and transmit/receive modules (T/R modules **128**). Each of the MCM **126** may include one of the RFIC **104** and four of the T/R modules **128**. In the example where the AESA **100** include a sixteen of the RFIC **104**, the AESA **100** may include sixteen of the MCM **126** each including one of the sixteen of the RFIC **104** and may include sixty-four of the T/R modules **128**, where each one of the sixteen of the MCM **126** includes four of the sixty-four of the T/R modules **128**.
(41) The controller **102** may be an array controller. The controller **102** may cause the each of the RFIC **104** within the MCM **126** to set, monitor, and adjust power levels for each of the T/R module **128** within the MCM **126**.
(42) The RFIC **104** may be coupled to the T/R modules **128**. The RFIC **104** may be coupled to the T/R modules **128** using one or more analog signals.
(43) The RFIC **104** may also be referred to as a RFIC transmit/receive (T/R) module chip. The RFIC **104** may be a mixed signal T/R RFIC module. The RFIC **104** may be a mixed signal T/R RFIC module in that the RFIC **104** may communicate using the digital logic signals over the primary point-to-point serial interfaces **106** and secondary point-to-point serial interfaces **108** and may communicate with the T/R modules **128** using the analog signals.
(44) The RFIC **104** may be configured to control the T/R modules **128**. For example, the RFIC **104** may be configured to provide bias and control of the T/R modules **128**. The RFIC **104** may be a subarray controller for the T/R modules **128**. The RFIC **104** may control the T/R modules **128**. The RFIC **104** may set, monitor, and adjust power levels for each of the T/R module **128** within the MCM **126**. The RFIC **104** may also be a beamformer. The RFIC **104** may cause the T/R modules **128** to form one or more beams.
(45) A chip count in the MCM **126** may be minimized using the RFIC **104**. The MCM **126** may include one of the RFIC **104** which controls four of the T/R modules **128**. The RFIC **104** may include one or more channels for the T/R modules **128**. For example, the RFIC **104** may include four channels, one for each of the T/R modules **128**. The RFIC **104** may channelize the T/R modules **128**. Channelizing the T/R modules **128** may allow the RFIC **104** to independently steer beams from each of the T/R modules **128**. The RFIC **104** may cause the T/R modules **128** to independently steer the beams at different frequencies. The RFIC **104** may independently provide phase adjustments and gain control for each of the T/R modules **128**.
(46) The T/R modules **128** of the MCM **126** may form a subarray of the AESA **100**. For example, the T/R modules **128** may form a two-by-two subarray within the MCM **126**. In the example where the AESA **100** include a four-by-four array of the MCM **126**, the AESA **100** may include the four-by-four array of the MCM **126** each with the two-by-two subarray of the T/R modules **128** within the MCM **126**. In this example, the AESA **100** may include an eight-by-eight array of the T/R modules **128** within the AESA **100**, for a total of sixty-four of the T/R modules **128**.
(47) The T/R modules **128** may include radiating elements which transmit and/or receive radio frequency (“RF”) signals, a ground plane, a thermal management layer, and the like. The radiating elements may include horizontal polarization elements and/or vertical polarization elements. For example, each of the T/R modules **128** may include one of the horizontal polarization elements and one of the vertical polarization elements (i.e., a dual polarization element). Dual (or more) polarizations for a given frequency may be provided (e.g., polarimetric radar). The T/R modules **128** may control the power, frequency, phase, time delay, and the like of the radiating elements. The T/R modules **128** may amplify the signals into/out of the radiating elements. The T/R modules **128** may also switch the radiating elements between transmitting and receiving the signals. The spacing of the RFIC **104** in the lattice may maintain the radiating elements at a one-half wavelength spacing.
(48) The RFIC **104** may include one or more semiconductor components. The RFIC **104** may include a transmit/receive (T/R) switch configured to alternate the RFIC **104** between a transmit

mode and receive mode. The RFIC **104** may include a power amplifier (PA) and low noise amplifier (LNA) coupled to the T/R switch. The RFIC **104** may alternate between PA operation and LNA operation via the T/R switch. The RFIC **104** may include a directional coupler to sense a power output of the PA. The RFIC **104** may include a limiter (e.g., a Schottky diode limiter/switch) to limit a power input to the LNA.

(49) The RFIC **104** may be made of a select material, such as, but not limited to, silicon germanium (SiGe). Silicon germanium may be a high volume, low cost, mixed signal digital analog RF material. Silicon germanium may be used because of a high circuit density (low cost per square millimeter).

(50) The T/R modules **128** may be radio frequency chain circuits. The T/R modules **128** may include one or more electronic components, such as, but not limited to, a T/R switch, a directional coupler, a PIN diode limiter, a power amplifier (PA), a low-noise amplifier (LNA), and the like. high power transmit amplifiers (e.g., a final power amplifier), duplexers, filters, low-noise receive amplifiers (e.g., an initial power amplifier), phase shifters, time delay units, transmit/receive switches, and the like. The T/R modules **128** may be multi-function chips, including a power amplifier and low noise amplifier.

(51) The T/R modules **128** may be made of a select material, such as, but not limited to, gallium arsenide (GaAs), gallium nitride (GaN), Indium Phosphide (InP), or the like.

(52) The RFIC **104** may include a receiver **130**, decoder **132**, a receiver **134**, a decoder **136**, router **138**, device registers **140**, an encoder **142**, a transmitter **144** (XMIT), an encoder **146**, a transmitter **148**, and the like.

(53) The receiver **130** and the receiver **134** may receive serial input from the primary point-to-point serial interfaces **106** and secondary point-to-point serial interfaces **108**, respectively. The receiver **130** and the receiver **134** may include a matching number of pins as the primary point-to-point serial interfaces **106** and secondary point-to-point serial interfaces **108**, respectively. For example, the receiver **130** and the receiver **134** may each include one pin where the network messages **110** are Manchester encoded. By way of another example, the receiver **130** and the receiver **134** may each include two pin, one pin for the clock signal **116** and a second pin for the data signal **118**. The receiver **130** and the receiver **134** may receive the serial input via the pins.

(54) The receiver **130** and/or the receiver **134** may be configured to perform clock and data recovery on the input. For example, the receiver **130** and/or the receiver **134** may be configured to recover the clock signal **116** and/or the data signal **118** from the input received. The clock and data recover may be required where the network messages **110** is encoded using Manchester encoding. The receiver **130** and/or the receiver **134** may not perform clock and data recovery on the input where the receiver **130** and the receiver **134** may each include two pins.

(55) The decoder **132** and the decoder **136** may also be referred to as serial decoders. The decoder **132** and the decoder **136** may be coupled to the receiver **130** and the receiver **134**, respectively. The decoder **132** and the decoder **136** may receive from the receiver **130** and the receiver **134**, respectively, the clock signal **116** and/or the data signal **118**. The decoder **132** and the decoder **136** may decode the data signal **118** from the receiver **130** and the receiver **134**, respectively. The decoder **132** and the decoder **136** may decode the data signal **118** to determine the frames (e.g., the command frame **112** and/or the data frames **114** of the network messages **110**).

(56) The router **138** may be coupled to the decoder **132** and/or the decoder **136**. The router **138** may receive the command frame **112** and/or the data frames **114** of the network messages **110**. The router **138** may receive the frames from the decoder **132** and the decoder **136** in parallel. The router **138** may discard the frames from the decoder **136** while the frames from the decoder **132** are valid.

(57) The router **138** may also be coupled to the device registers **140**, the encoder **142**, and/or the encoder **146**. The router **138** may route the command frame **112** and/or the data frames **114** of the network messages **110** to the device registers **140**, the encoder **142**, and/or the encoder **146**. The router **138** may route the command frame **112** and/or the data frames **114** of the network messages

110 to the device registers **140**, the encoder **142**, and/or the encoder **146** based on the type of the command frame **112**, as will be described further herein. For example, the routing of the command frame **112** and/or the data frames **114** may vary between the read command frame **112***a*, write command frame **112***b*, the write-all command frame **112***c*, and/or the sync command frame **112***d*.

(58) The device registers **140** may receive the command frame **112** and/or the data frames **114** of the network messages **110**. The device registers **140** may maintain one or more portions of the command frame **112** and/or the data frames **114** in the device registers **140**. The device registers **140** may provide control and/or data storage/memory distribution among the MCM **126**. The local memory within the MCM **126** may allow for the T/R modules **128** to switch the beam more quickly than storing data within the controller **102**.

(59) The device registers **140** may include a bit length. The bit length may be a multiple of eight (e.g., one byte). For example, the device registers **140** may be 8-bit registers, 16-bit registers, 24-bit registers, 32-bit registers, 64-bit registers, or the like. In embodiments, the device registers **140** are 16-bit registers.

(60) One or more of the device registers **140** may be beam control registers. The device registers **140** may include the beam control registers for receiving beam control commands. The RFIC **104** may use the beam control commands to control the beam (e.g., polarity, gain, phase) of the T/R modules **128**. Thus, the beam of the T/R modules **128** may be controlled using the beam control commands stored in the device registers **140**. The device registers **140** may include at least one of the beam control registers for each of the T/R modules **128**. The device registers **140** may allow for localized storage of the beam control commands. The computational burden to drive the beams may be distributed to the RFIC **104** from the controller **102**.

(61) The encoder **142** and/or the encoder **146** may also be referred to as serial encoders. The encoder **142** and/or the encoder **146** may be coupled to the router **138**. The encoder **142** and/or the encoder **146** may receive the command frame **112** and/or the data frames **114** of the network messages **110** from the router **138**. The encoder **142** and/or the encoder **146** may encode the command frame **112** and/or the data frames **114** of the network messages **110** received from the router **138** into the clock signal **116** and/or the data signal **118** for the primary point-to-point serial interfaces **106** and/or the secondary point-to-point serial interfaces **108**, respectively.

(62) The transmitter **144** and/or the transmitter **148** may be coupled to the encoder **142** and/or the encoder **146**, respectively. The transmitter **144** and/or the transmitter **148** may be configured to output to the primary point-to-point serial interfaces **106** and/or the secondary point-to-point serial interfaces **108**, respectively, the clock signal **116** and/or the data signal **118**.

(63) The transmitter **144** and/or the transmitter **148** may include a matching number of pins as the primary point-to-point serial interfaces **106** and secondary point-to-point serial interfaces **108**, respectively.

(64) The controller **102** and/or the RFIC **104** may be configured to transmit network messages **110** between the controller **102** and the RFIC **104** over the primary point-to-point serial interfaces **106** and/or the secondary point-to-point serial interfaces **108**. The controller **102** and/or the RFIC **104** may be configured to transmit the network messages **110** between the controller **102** and the RFIC **104** over the primary point-to-point serial interfaces **106** and/or the secondary point-to-point serial interfaces **108** using logic signals. The controller **102** and/or the RFIC **104** may transmit the logic signals on the primary point-to-point serial interfaces **106** and the secondary point-to-point serial interfaces **108**.

(65) The primary point-to-point serial interfaces **106** and/or the secondary point-to-point serial interfaces **108** may include two logic signals. For example, the primary point-to-point serial interfaces **106** and/or the secondary point-to-point serial interfaces **108** may include only a clock signal **116** and a data signal **118**. The clock signal **116** may provide transitions for each serial data bit in the network messages **110**. The data signal **118** may include serial data bits.

(66) The primary point-to-point serial interfaces **106** and/or the secondary point-to-point serial

interfaces **108** may include one or more pins. The primary point-to-point serial interfaces **106** and/or the secondary point-to-point serial interfaces **108** may each include a single pin (as depicted). The clock signal **116** may be encoded onto the data signal **118** of the network messages **110** using Manchester encoding, allowing both the clock signal **116** and the data signal **118** to be carried by the single pin. The Manchester encoding may remove clock skew. It is further contemplated that the primary point-to-point serial interfaces **106** and/or the secondary point-to-point serial interfaces **108** may each include two pins (not depicted). A first of the two pins may carry the clock signal **116** and a second of the two pins may carry the data signal **118**. The clock signal **116** and a data signal **118** of the network messages **110** are not encoded onto the two pins using Manchester encoding where the first of the two pins carries the clock signal **116** and the second of the two pins carries the data signal **118**.
(67) The primary point-to-point serial interfaces **106** and/or the secondary point-to-point serial interfaces **108** may use timeout-framing. The timeout-framing may include staring a frame start after a timeout period without receiving the clock signal **116** and/or the data signal **118**.
(68) Thus, the controller **102** may include sixteen pins when using Manchester encoding and where the AESA **100** include a four-by-four array of the RFIC **104**. For example, the controller **102** may include four pin outs for the output to the primary point-to-point serial interfaces **106**, four pin outs for the output to the secondary point-to-point serial interfaces **108**, four pin ins for the input from the primary point-to-point serial interfaces **106**, and four pin ins for the input from the secondary point-to-point serial interfaces **108**. This arrangement may reduce the pins of the controller **102** as compared to the multi-drop SPI bus arrangement for a four-by-four array of the RFIC **104**.
(69) The primary point-to-point serial interfaces **106** and/or the secondary point-to-point serial interfaces **108** may not include a chip select logic signal (also referred to as a sub select logic signal). Thus, the primary point-to-point serial interfaces **106** and/or the secondary point-to-point serial interfaces **108** do not include a separate logic signal for indicating the address of the network messages **110**.
(70) The network messages **110** may follow a serial protocol. The network messages **110** may use positional addressing to define the address of the data frames **114**. For example, the read network messages **110***a* and/or the write network messages **110***b* may use positional addressing to define the address of the data frames **114** within the read network messages **110***a* and/or within the write network messages **110***b*. The position of the data frames in the network messages **110** may indicate the address for the data frames. The RFIC **104** may be addressed using the position of the data frames **114** in the network messages **110**. The address may refer to which of the RFIC **104** the data frames are to and/or from. The address may be determined by the position of the RFIC **104** in the grid network. The positional addressing may allow dense and larger number of the RFIC **104** in the AESA **100** with less signals and faster clock speeds. The positional addressing may not require the use of a chip select logic signal.
(71) The network messages **110** may include one or more frames (e.g., command frame **112**, data frames **114**). The frames may refer to a digital data transmission unit. The frames may be transmitted over the serial data pins. The network messages **110** may include a command frame **112** followed by data frames **114**. The first frame in the network messages **110** may be the command frame **112**. A remainder of frames in the network messages **110** may be the data frames **114**. The data frames **114** may immediately follow the command frame **112**. For example, the data frames **114** of the read network messages **110***a* may immediately follow the read command frame **112***a*, the data frames **114** of the write network messages **110***b* may immediately follow the write command frame **112***b*, and/or the data frames **114** of the write-all network messages **110***c* may immediately follow the write-all command frame **112***c*.
(72) The network messages **110** may include a select number of the data frames **114**. The number of the data frames **114** may be based on the number of the RFIC **104** in the rows of the grid network, in the columns of the grid network, and/or whether the network messages **110** are

transmitted over the primary point-to-point serial interfaces **106** and/or the secondary point-to-point serial interfaces **108**. The number of the data frames **114** in the network messages **110** transmitted over the primary point-to-point serial interfaces **106** may be equal to the number of the RFIC **104** in the row. The number of the data frames **114** in the network messages **110** transmitted over the secondary point-to-point serial interfaces **108** may be equal to the number of the RFIC **104** in the column.

(73) The current of the network messages **110** may terminate after a frame period without additional frames or with an idle pattern. A subsequent of the network messages **110** may be transmitted after the current of the network messages **110** is terminated.

(74) The command frame **112** (CMD) may also be referred to as control frames (CTL).

(75) The network messages **110** may be read network messages **110***a*, write network messages **110***b*, write-all network messages **110***c*, or sync network messages **110***d*. The command frame **112** may include a read command frame **112***a*, a write command frame **112***b*, a write-all command frame **112***c*, and/or a sync command frame **112***d*. The network messages **110** may be the read network messages **110***a*, the write network messages **110***b*, the write-all network messages **110***c*, or the sync network messages **110***d* where the network messages **110** includes the read command frame **112***a*, the write command frame **112***b*, the write-all command frame **112***c*, or the sync command frame **112***d*, respectively.

(76) Each of the frames (e.g., the command frame **112** (e.g., the read command frame **112***a*, the write command frame **112***b*, the write-all command frame **112***c*, or the sync command frame **112***d*), the data frames **114**) may include a same frame size. The frame size may also be referred to as bit length or the number of bits within the frames. For example, the read command frame **112***a*, the write command frame **112***b*, the write-all command frame **112***c*, the sync command frame **112***d*, and/or the data frames **114** may each include a same frame size. Each of the frames may include the same frame size to enable positional addressing within the AESA **100**.

(77) Each of the frames may include a start bit **120**, a header **122**, and a packet payload **124**.

(78) The start bit **120** may indicate the start of the frame. The start bit **120** may provide a one-bit gap between consecutive of frames. For example, the start bit **120** may provide a one-bit gap between consecutive of frames within one of the network messages **110**. By way of another example, the start bit **120** may provide a one-bit gap between a last frame within a current of the network messages **110** and a first frame within a subsequent of the network messages **110**.

(79) The header **122** may follow the start bit **120**. The header **122** may also be referred to as command bits. The header **122** may indicate whether the frame is the data frames **114**, the read command frame **112***a*, the write command frame **112***b*, the write-all command frame **112***c*, or the sync command frame **112***d*. The header **122** may include a plurality of bits which indicate whether the frame is the data frames **114**, the read command frame **112***a*, the write command frame **112***b*, the write-all command frame **112***c*, or the sync command frame **112***d*. For example, the header **122** may include a first bit, a second bit, and a third bit.

(80) The first bit of the header **122** may indicate whether the frame is the command frame **112** or the data frames **114**. For example, the first bit may be one to indicate the frame is the data frames **114** and may be zero to indicate the frame is the command frame **112**. The specific values of zero and one for first bit of the header **122** are exemplary only and are not intended to be limiting. The meaning of zero and one may be reversed while enabling the first bit of the header **122** to indicate whether the frame is the command frame **112** or the data frames **114**.

(81) The second and third bit of the header **122** may indicate whether the command frame **112** is the read command frame **112***a*, the write command frame **112***b*, the write-all command frame **112***c*, or the sync command frame **112***d*. For example, the second bit and third bit may be zero-zero to indicate the command frame **112** is the read command frame **112***a*, may be zero-one to indicate the command frame **112** is the write command frame **112***b*, may be one-zero to indicate the command frame **112** is the write-all command frame **112***c*, or may be one-one to indicate the command frame

112 is the sync command frame **112**_d_. The specific values of zero and one for second and third bits of the header **122** are exemplary only and are not intended to be limiting. It is contemplated that the second and third bit of the header **122** may include four permutations of the zero and one which each enable the second and third bit of the header **122** to indicate whether the command frame **112** is the read command frame **112**_a_, the write command frame **112**_b_, the write-all command frame **112**_c_, or the sync command frame **112**_d_.
(82) Thus, the specific implementation of zero and ones for the first three bits of the header **122** of the frames may vary and is intended to be limiting so long as the first bit indicates whether the frame is the command frame **112** or the data frames **114** and the second and third bit indicate whether the command frame **112** is the read command frame **112**_a_, the write command frame **112**_b_, the write-all command frame **112**_c_, or the sync command frame **112**_d_.
(83) The packet payload **124** may follow the header **122**. The packet payload **124** may include data bits (D), address bits (A), and/or sync bits(S). The bits (e.g., the data bits (D), address bits (A), and/or sync bits(S)) of the packet payload **124** may be based on the type of the frame. The packet payload **124** of the data frames **114** may include the data bits (D). The packet payload **124** of the read command frame **112**_a_, the write command frame **112**_b_, and/or the write-all command frame **112**_c_ may include the address bits (A). The packet payload **124** of the sync command frame **112**_d_ may include the sync bits(S).
(84) The data frame **114** may be used for reading and writing to the device registers **140**. The packet payload **124** of the data frames **114** may be written and/or read from the device registers **140** of the RFIC **104**. When writing to the device registers **140**, the data bits (D) may specify a polarity, gain, phase, or the like for the T/R modules **128**. When reading from the device registers **140**, the data bits (D) may specify the signal received from the T/R modules **128** (e.g., the radio frequency signal received from the T/R modules **128**).
(85) The data frames **114** may use the second bit and the third bit of the header **122** for the data bits (D) of the packet payload **124**. Thus, the packet payload **124** of the data frames **114** may be two bits larger than the packet payload **124** of the command frame **112**. Using the second and third bits of the header **122** for the packet payload **124** may increase the data rate of the network messages **110** without increasing the length of the frames.
(86) The address bits (A) of the read command frame **112**_a_, the write command frame **112**_b_, and/or the write-all command frame **112**_c_ may contain the register address of one or more of the device registers **140** of the RFIC **104**. However, the address bits (A) of the read command frame **112**_a_, the write command frame **112**_b_, and/or the write-all command frame **112**_c_ may not indicate to/from which of the RFIC **104** the network messages **110** are addressed.
(87) The sync bits(S) of the sync command frame **112**_d_ may propagate discrete values through the RFIC **104**. The sync bits (B) may be a discrete bit that is sent to each of the RFIC **104**. A remainder is left for the discrete values. The discrete values may also be referred to as control bits. The discrete values may turn on and/or off one or more functions of the RFIC **104**.
(88) Each of the frames (e.g., the command frame **112**, the data frames **114**) may include a select number of bits. The number of bits in each of the frames may be based on the length of the device registers **140** within the RFIC **104**. The length of the frames may be selected such that the packet payload **124** of the data frames **114** match the length of the device registers **140**. The length of the packet payload **124** of the data frame **114**, including the including the two additional bits from the header **122**, may be in multiples of eight. For example, the device registers **140** may be sixteen-bit registers. The length of each of the frames may be eighteen bits. The bits of the data frames **114** may include the one bit for the start bit **120**, the one bit for the header **122**, and sixteen data bits (D**15** through D**0**) for the packet payload **124**. The bits of the read command frame **112**_a_, the write command frame **112**_b_, and/or the write-all command frame **112**_c_ may include the one bit for the start bit **120**, the three bits for the header **122**, and fourteen address bits (A**13** through A**0**) for the packet payload **124**. The bits of the sync command frame **112**_d_ may include the one bit for the start

bit **120**, the three bits for the header **122**, and fourteen sync bits (S**13** through S**0**) for the packet payload **124**.

(89) The packet payload **124** may also include one or more parity bits (P). For example, the parity bits (P) may be a last bit of the packet payload **124** (e.g., D**0**, A**0**, S**0**). The parity bits (P) may provide error correcting code (ECC) to the frames. For example, the bit-value of the parity bits (P) may be selected such that a Hamming weight of the frame may always be even or always be odd.

(90) The read network messages **110***a* may be used to read different of the packet payloads **124** from the same register address of each of the RFIC **104**. The read network messages **110***a* initially transmitted by the controller **102** may include zero of the data frames **114**. The position of the data frames **114** in the read network messages **110***a* may indicate the address of the data frames **114** to the RFIC **104**. For example, the position of the data frames **114** from first to last may correspond to the position of the RFIC **104** from first to last along the row and/or along the column. The read command frame **112***a* may include a register address of one or more of the device registers **140** of the RFIC **104**. The read command frame **112***a* may cause the RFIC **104** to initiate read operations. In response to receiving the read command frame **112***a*, the router **138** may retrieve data which is maintained at the register address of one or more of the device registers **140** of the RFIC **104**. The router **138** may then append the data which is maintained at the register address into the packet payload **124** of a new copy of the data frames **114** at the end of the network messages **110**. The router **138** may forward the network messages **110** with the new copy of the data frames **114** to the encoder **142** and/or the encoder **146**. The RFIC **104** may append the data frames **114** to the read network messages **110***a* in response to receiving the read command frame **112***a*. Thus, the number of the data frames **114** in the read network messages **110***a* may increase by one after being received by each of the RFIC **104** in series. The read network messages **110***a* received by the controller **102** may include M-number or N-number of the data frames **114**, where M or N is the number of the RFIC **104** along the rows and columns, respectively.

(91) For example, a first of the RFIC **104** may receive a serial data input (SDI_**1**) from the controller **102** with only the read command frame **112***a*. The router **138** of the first of the RFIC **104** may read the packet payload **124** of a first data frame **114-1** from the register address indicated in the read command frame **112***a* and append to the read network messages **110***a*. A second of the RFIC **104** may receive a serial data input (SDI_**2**) from the first of the RFIC **104** with the read command frame **112***a* and the first data frame **114-1** addressed in position from the first of the RFIC **104**. The router **138** of the second of the RFIC **104** may read the packet payload **124** of a second data frame **114-2** from the register address indicated in the read command frame **112***a* and append to the read network messages **110***a*. A third of the RFIC **104** may receive a serial data input (SDI_**3**) from the second of the RFIC **104** with the read command frame **112***a*, the first data frame **114-1** addressed in position from the first of the RFIC **104**, and the second data frame **114-2** addressed in position from the second of the RFIC **104**. The router **138** of the third of the RFIC **104** may read the packet payload **124** of a third data frame **114-3** from the register address indicated in the read command frame **112***a* and append to the read network messages **110***a*. A fourth of the RFIC **104** may receive a serial data input (SDI_**4**) from the third of the RFIC **104** with the read command frame **112***a*, the first data frame **114-1** addressed in position from the first of the RFIC **104**, the second data frame **114-2** addressed in position from the second of the RFIC **104**, and the third data frame **114-3** addressed in position from the third of the RFIC **104**. The router **138** of the fourth of the RFIC **104** may read the packet payload **124** of a fourth data frame **114-4** from the register address indicated in the read command frame **112***a* and append to the read network messages **110***a*. The fourth of the RFIC **104** may output a serial data output (SDO_**4**) to the controller **102** with the read command frame **112***a*, the first data frame **114-1** addressed in position from the first of the RFIC **104**, the second data frame **114-2** addressed in position from the second of the RFIC **104**, the third data frame **114-3** addressed in position from the third of the RFIC **104**, and the fourth data frame **114-4** addressed in position from the fourth of the RFIC **104**.

(92) The write network messages **110***b* may be used to write different of the packet payloads **124** to the same register address of each of the RFIC **104**. The write network messages **110***b* initially transmitted by the controller **102** may include M-number or N-number of the data frames **114**, where M or N is the number of the RFIC **104** along the rows and columns, respectively. The position of the data frames **114** in the write network messages **110***b* may indicate the address of the data frames **114** to the RFIC **104**. For example, the position of the data frames **114** from first to last may correspond to the position of the RFIC **104** from first to last along the row and/or along the column. The write command frame **112***b* may include a register address of one or more of the device registers **140** of the RFIC **104**. The write command frame **112***b* may cause the RFIC **104** to initiate write operations. In response to receiving the write command frame **112***b*, the router **138** may write the packet payload **124** from the first of the data frames **114** in the network messages **110** to the register address of the device registers **140**. The router **138** may also truncate the first of the data frames **114** from the write network messages **110***b* and forward the write command frame **112***b* and the remainder of the data frames **114** to the encoder **142** and/or the encoder **146**. The RFIC **104** may truncate the data frames **114** from the write network messages **110***b* in response to receiving the write command frame **112***b*. Thus, the number of the data frames **114** in the write network messages **110***b* may decrease by one after being received by each of the RFIC **104** in series. The write network messages **110***b* received by the controller **102** may include zero of the data frames **114**.
(93) For example, the first of the RFIC **104** may receive a serial data input (SDI_**1**) from the controller **102** with the write command frame **112***b*, the first data frame **114-1** addressed in position to the first of the RFIC **104**, the second data frame **114-2** addressed in position to the second of the RFIC **104**, the third data frame **114-3** addressed in position to the third of the RFIC **104**, and the fourth data frame **114-4** addressed in position to the fourth of the RFIC **104**. The router **138** of the first of the RFIC **104** may write the packet payload **124** of the first data frame **114-1** to the register address indicated in the write command frame **112***b* and truncate the first data frame **114-1** from the write network messages **110***b*. The second of the RFIC **104** may receive a serial data input (SDI_**2**) from the first of the RFIC **104** with the write command frame **112***b*, the second data frame **114-2** addressed in position to the second of the RFIC **104**, the third data frame **114-3** addressed in position to the third of the RFIC **104**, and the fourth data frame **114-4** addressed in position to the fourth of the RFIC **104**. The router **138** of the second of the RFIC **104** may write the packet payload **124** of the second data frame **114-2** to the register address indicated in the write command frame **112***b* and truncate the second data frame **114-2** from the write network messages **110***b*. The third of the RFIC **104** may receive a serial data input (SDI_**3**) from the second of the RFIC **104** with the write command frame **112***b*, the third data frame **114-3** addressed in position to the third of the RFIC **104**, and the fourth data frame **114-4** addressed in position to the fourth of the RFIC **104**. The router **138** of the third of the RFIC **104** may write the packet payload **124** of the third data frame **114-3** to the register address indicated in the write command frame **112***b* and truncate the third data frame **114-3** from the write network messages **110***b*. The fourth of the RFIC **104** may receive a serial data input (SDI_**4**) from the third of the RFIC **104** with the write command frame **112***b* and the fourth data frame **114-4** addressed in position to the fourth of the RFIC **104**. The router **138** of the fourth of the RFIC **104** may write the packet payload **124** of the fourth data frame **114-4** to the register address indicated in the write command frame **112***b* and truncate the fourth data frame **114-4** from the write network messages **110***b*. The fourth of the RFIC **104** may output a serial data output (SDO_**4**) to the controller **102** with only the write command frame **112***b*.
(94) The RFIC **104** may each write all the packet payloads **124** from the data frames **114** of the write-all network messages **110***c*. The write-all network messages **110***c* may be used to write the same of the packet payloads **124** to the same register address of each of the RFIC **104**. The write-all network messages **110***c* may be used to initialize the RFIC **104**. The write-all network messages **110***c* initially transmitted by the controller **102** may include any integer number of the data frames

114. The position of the data frames **114** in the write-all network messages **110***c* may not indicate the address of the data frames **114** to the RFIC **104**. Instead, the data frames **114** in the write-all network messages **110***c* may be addressed to each of the RFIC **104**. The write-all command frame **112***c* may include a register address of one or more of the device registers **140** of the RFIC **104**. The write-all command frame **112***c* may cause the RFIC **104** to initiate write-all operations. The router **138** may forward the data frames **114** to the register address indicated by the write-all command frame **112***c*. The router **138** may also forward the write-all network messages **110***c* without changing any of the data frames **114** to the encoder **142** and/or the encoder **146**. Thus, each of the RFIC **104** may write the same information using the write-all network messages **110***c*. The write-all network messages **110***c* received by the controller **102** may include the same number of the data frames **114** as was initially transmitted.
(95) The RFIC **104** may auto-increment register address indicated in the write-all command frame **112***c* when writing the packet payload **124** of multiple of the data frame **114**. Thus, the packet payload **124** of multiple of the data frame **114** may be written to consecutive registers within the device registers **140**.
(96) For example, the first of the RFIC **104** may receive a serial data input (SDI_**1**) from the controller **102**, the second of the RFIC **104** may receive a serial data input (SDI_**2**) from the first of the RFIC **104**, the third of the RFIC **104** may receive a serial data input (SDI_**3**) from the second of the RFIC **104**, the fourth of the RFIC **104** may receive a serial data input (SDI_**4**) from the third of the RFIC **104**, and the fourth of the RFIC **104** may output a serial data output (SDO_**4**) to the controller **102**, where each of the serial data inputs and serial data outputs include the write-all command frame **112***c* and a matching number of the data frames **114**. As depicted, each of the serial data inputs and serial data outputs include first data frame **114-1** through fourth data frame **114-4**, although this number of data frames is not intended to be limiting. The router **138** of each of the first, second, third, and fourth of the RFIC **104** may write the packet payload **124** of the first data frame **114-1** at the register address indicated in the write-all command frame **112***c*, the packet payload **124** of the second data frame **114-2** at the register address indicated in the write-all command frame **112***c* incremented by one, the packet payload **124** of the third data frame **114-3** at the register address indicated in the write-all command frame **112***c* incremented by two, and the packet payload **124** of the fourth data frame **114-4** at the register address indicated in the write-all command frame **112***c* incremented by three.
(97) The sync network messages **110***d* may not include any of the data frames **114**. The sync command frame **112***d* may cause the RFIC **104** to initiate sync operations. The sync command frame **112***d* may include one or more discrete values. The router **138** may forward the sync network messages **110***d* to the encoder **142** and/or the encoder **146**. The sync network messages **110***d* may be transmitted with minimal latency due to not including any of the data frames **114**. The sync network messages **110***d* may interrupt current of the network messages **110** to synchronize each of the RFIC **104**.
(98) For example, the first of the RFIC **104** may receive a serial data input (SDI_**1**) from the controller **102**, the second of the RFIC **104** may receive a serial data input (SDI_**2**) from the first of the RFIC **104**, the third of the RFIC **104** may receive a serial data input (SDI_**3**) from the second of the RFIC **104**, the fourth of the RFIC **104** may receive a serial data input (SDI_**4**) from the third of the RFIC **104**, and the fourth of the RFIC **104** may output a serial data output (SDO_**4**) to the controller **102**, where each of the serial data inputs and serial data outputs include the sync command frame **112***d*.
(99) One or more frames of the read network messages **110***a*, the write network messages **110***b*, and/or the write-all network messages **110***c* may overlap when transmitting the frames between the controller **102** and the RFIC **104**. For example, the RFIC **104** may forward the command frames **112** and/or the data frames **114** while receiving the network messages **110**. The sync command frames **112***d* of the sync network messages **110***d* may not overlap when transmitting the frames

between the controller **102** and the RFIC **104**.

(100) It is contemplated that the AESA **100** may achieve 44/88 Mbps data rate (clockless/clocked) at a clock rate of one-hundred MHz. For example, each of the frame may be eighteen-bits. The signal may have two transitions of the clock for each one bit using Manchester encoding (e.g., fifty MHz). The packet payload **124** of each of the data frames **114** may be sixteen bits, resulting in a speed of 44 Mbps for Manchester encoding.

(101) The AESA **100** may use the read network messages **110***a*, the write network messages **110***b*, the write-all network messages **110***c*, or the sync network messages **110***d* for the primary input/outputs on the primary point-to-point serial interfaces **106** and/or as the secondary input/outputs on the secondary point-to-point serial interfaces **108**. The primary inputs may be passed through to the primary output on the primary point-to-point serial interfaces **106.** The secondary inputs may be passed through to the secondary outputs on the secondary point-to-point serial interfaces **108**.

(102) The secondary point-to-point serial interfaces **108** may allow for continued operation and control in the event of a single point failure on the primary point-to-point serial interfaces **106**. If the primary point-to-point serial interfaces **106** providing the primary input to one or more of the RFIC **104** fail, then the secondary point-to-point serial interfaces **108** providing the secondary input to one or more of the RFIC **104** may be used to reroute the frames between the RFIC **104**. Broken or failed of the RFIC **104** and/or the primary point-to-point serial interfaces **106** may be remapped using the secondary point-to-point serial interfaces **108** while maintaining communication of the network messages **110** between the controller **102** and the RFIC **104**. Thus, the secondary point-to-point serial interfaces **108** may provide fault-tolerance and prevent the RFIC **104** in a same row or column from being inaccessible due to failure of single of the RFIC **104** or a single of the primary point-to-point serial interfaces **106**.

(103) FIG. **2** depicts a flow diagram of a method **200**, in accordance with one or embodiments of the present disclosure. The method **200** may be executed by the controller **102** and/or the RFIC **104** of the AESA **100**. The method **200** may be a method for monitoring the health of the RFIC **104**.

(104) In a step **210**, the controller **102** causes the RFIC **104** to write a health status message into the device registers **140** during initialization of the AESA **100**. The controller **102** may cause the RFIC **104** to write the health status message into the device registers **140** using the write network messages **110***b*. The write command frame **112***b* of the write network messages **110***b* may include a health status register address of the device registers **140**. The health status register address may also be referred to as a scratchpad register address. Each of the RFIC **104** may include a unique value for the health status message. For example, the health status message may include the number of sequence of the RFIC **104** along the row of the primary point-to-point serial interfaces **106** and/or the number of the sequence of the RFIC **104** along the column of the secondary point-to-point serial interfaces **108**. The location of the device registers **140**. The health status message may also indicate that the RFIC **104** is currently receiving the primary inputs from the primary point-to-point serial interfaces **106**.

(105) In a step **220**, the controller **102** transmits a same of the network messages **110** on the primary point-to-point serial interfaces **106** and the secondary point-to-point serial interfaces **108**.

(106) In a step **230**, the RFIC **104** detects whether the primary input from the primary point-to-point serial interfaces **106** and/or the secondary input from the secondary point-to-point serial interfaces **108** are or are not operational. The RFIC **104** may detect whether the primary input and/or the secondary input are or are not operational based on the parity bits, a missing clock signal, the idle pattern of the network messages **110**, or the like. The RFIC **104** may default to the primary input from the primary point-to-point serial interfaces **106** when the primary input is operational. For example, the RFIC **104** may default to the primary input as a source of register/control data.

(107) In a step **240**, the RFIC **104** may update the health status message in the device registers **140**

and revert to using the secondary input from the secondary point-to-point serial interfaces **108** in response to detecting the primary input is not operational. For example, the RFIC **104** may update the health status message in the device registers **140** to indicate that the primary input has failed and is not being received. The RFIC **104** may switch to the secondary input when the primary input is not operational. The secondary input may bypass previous of the RFIC **104** and/or previous of the primary point-to-point serial interfaces **106**.

(108) In a step **250**, the controller **102** may read the health status message. The controller **102** may read the health status message from the RFIC **104** by sending the read network messages **110***a* with the read command frame **112***a* including the packet payload **124** with the address at which the health status message is stored. The RFIC **104** may append the data frames **114** with the updated health status message. The controller **102** may receive the read network messages **110***a* with the data frames **114** appended to include the updated health status message and determine which of the RFIC **104** are operational and which of the RFIC **104** are using the primary inputs vs the secondary inputs. Thus, the controller **102** may monitor the health of the AESA **100** by sending the read network messages **110***a* to read the health status message from the device registers **140**. The controller **102** may monitor the health of the AESA **100** by measuring the number of the data frames **114** received in combination with the health status message in the packet payload **124** of the data frames **114**.

(109) In a step **260**, the controller **102** may update a network status and remap the network messages around defective of the RFIC **104**. The controller **102** may determine defective of the RFIC **104** based on the health status messages. The network status may indicate the RFIC **104** are active or defective. The controller **102** may remaps the network messages **110** based on which of the RFIC **104**, primary point-to-point serial interfaces **106**, and/or secondary point-to-point serial interfaces are operational and which are not.

(110) FIG. **3** depicts the AESA **100**, in accordance with one or more embodiments of the present disclosure. The AESA **100** may include the RFIC **104** may be arranged in a plurality of arrays. For example, the AESA **100** may include two or more arrays of the RFIC **104**. Each of the arrays may include the lattice of the RFIC **104**. For example, the arrays of the RFIC **104** may be arranged in a four-by-four array (e.g., a four-by-four array of the MCM **126**). The primary point-to-point serial interfaces **106** and secondary point-to-point serial interfaces **108** may connect the arrays. The primary point-to-point serial interfaces **106** and secondary point-to-point serial interfaces **108** may connect between the arrays before going back to the controller **102**. Thus, the primary point-to-point serial interfaces **106** and secondary point-to-point serial interfaces **108** may span between adjacent arrays.

(111) Referring generally again to the figures.

(112) The AESA **100** may be a radar system. The radar system may be an avionic radar system, a ground-based radar system, a space-based radar system, a naval-based radar system, or the like. The radar system may be a weather radar system, a sense and avoid (S&A) radar system, a ground moving target indicator (GMTI) radar system, a search and rescue radar system, a “brown out” radar system, a multimode signal intelligence radar system, an electronic warfare radar system, a border surveillance radar system, a fire control radar system, a millimeter wave (MMW) imaging and landing radar system, or the like. The radar system may be a hybrid frequency modulated continuous wave (FMCW)/pulsed radar system. The AESA **100** may be used in radar, communication, navigation, and surveillance systems (CNS), including Satcom. The AESA **100** may be a detect-and-avoid or due regard radar (DRR) system, a maritime rotary wing search-and-rescue radar system, a degraded visual environment (DVE) radar imaging system.

(113) The AESA **100** may be a low probability of intercept/low probability of detection (LPI/LPD) high data rate (HDR) directional link systems, e.g., for man packs, ground vehicles, maritime and airborne applications, satellite-on-the-move (SOTM) communications, or special operations. The AESA **100** may be an anti-access area denial (A2AD) burn-through jammer AESA systems.

(114) The controller **102** may be configured to provide low voltage differential signaling (LVDS) for the primary point-to-point serial interfaces **106** and/or the secondary point-to-point serial interfaces **108** between the controller **102** and the RFIC **104** and/or the primary point-to-point serial interfaces **106** and/or the secondary point-to-point serial interfaces **108** between the RFIC **104** and the controller **102**. The LVDS may be a differential interface which may allow the primary point-to-point serial interfaces **106** and/or the secondary point-to-point serial interfaces **108** to go longer distances while maintaining signal integrity. The LVDS may be a difference in voltage between two of the wires, thereby cancelling the noise which each of the wires experience along the length of the primary point-to-point serial interfaces **106** and/or the secondary point-to-point serial interfaces **108**.
(115) The RFIC **104** may include one or more components for controlling the T/R modules **128**. For example, RFIC **104** may include: limiter T/R switching, Low Noise Amplifiers (LNA), Power Amplifier (PA), variable gain reviewer and exciter amplification, attenuator phase shift (or time delay), temperature sensing, power detection, phase detection, distributed computing and memory, power supply, and the like.
(116) The controller **102** may include one or more processors and/or memory (not depicted). The processors may be any device having one or more processing elements, which execute program instructions from memory. For example, the processors may include a multi-core processor, a single-core processor, a reconfigurable logic device (e.g., FPGAs), a digital signal processor (DSP), a special purpose logic device (e.g., ASICs), or other integrated formats. Aspects of the embodiments disclosed herein, in whole or in part, can be equivalently implemented in integrated circuits, as one or more computer programs running on one or more computers (e.g., as one or more programs running on one or more computer systems), as one or more programs running on one or more processors (e.g., as one or more programs running on one or more microprocessors), as firmware, or as virtually any combination thereof, and that designing the circuitry and/or writing the code for the software/and or firmware would be well within the skill of one skilled in the art in light of this disclosure. Such hardware, software, and/or firmware implementation may be a design choice based on various cost, efficiency, or other metrics. In this sense, the processor(s) may include any microprocessor-type device configured to execute software algorithms and/or instructions. In general, the term "processor" may be broadly defined to encompass any device having one or more processing elements, which execute program instructions from memory, from firmware, or by hardware implemented functions. It should be recognized that the steps described throughout the present disclosure may be carried out by the processors.
(117) A module can take the form of one or more analog circuits, electronic circuits (e.g., integrated circuits (IC), discrete circuits, system on a chip (SOCs) circuit, microcontrollers, etc.), telecommunication circuits, hybrid circuits, and any other type of "circuit." In this regard, the modules can include any type of component for accomplishing or facilitating achievement of the operations described herein. For example, a circuit as described herein can include one or more transistors, logic gates (e.g., NAND, AND, NOR, OR, XOR, NOT, XNOR, etc.), resistors, multiplexers, registers, capacitors, inductors, diodes, wiring, and so on), and programmable hardware devices (e.g., field programmable gate arrays, programmable array logic, programmable logic devices or the like). The modules can include a processor and one or more memory devices for storing instructions that are executable by each of the processors.
(118) One skilled in the art will recognize that the herein described components (e.g., operations), devices, objects, and the discussion accompanying them are used as examples for the sake of conceptual clarity and that various configuration modifications are contemplated. Consequently, as used herein, the specific exemplars set forth and the accompanying discussion are intended to be representative of their more general classes. In general, use of any specific exemplar is intended to be representative of its class, and the non-inclusion of specific components (e.g., operations), devices, and objects should not be taken as limiting.

(119) Those having skill in the art will appreciate that there are various vehicles by which processes and/or systems and/or other technologies described herein can be affected (e.g., hardware, software, and/or firmware), and that the preferred vehicle will vary with the context in which the processes and/or systems and/or other technologies are deployed. For example, if an implementer determines that speed and accuracy are paramount, the implementer may opt for a mainly hardware and/or firmware vehicle; alternatively, if flexibility is paramount, the implementer may opt for a mainly software implementation; or, yet again alternatively, the implementer may opt for some combination of hardware, software, and/or firmware. Hence, there are several possible vehicles by which the processes and/or devices and/or other technologies described herein may be affected, none of which is inherently superior to the other in that any vehicle to be utilized is a choice dependent upon the context in which the vehicle will be deployed and the specific concerns (e.g., speed, flexibility, or predictability) of the implementer, any of which may vary.
(120) The previous description is presented to enable one of ordinary skill in the art to make and use the invention as provided in the context of a particular application and its requirements. As used herein, directional terms such as “top,” “bottom,” “over,” “under,” “upper,” “upward,” “lower,” “down,” and “downward” are intended to provide relative positions for purposes of description, and are not intended to designate an absolute frame of reference. Various modifications to the described embodiments will be apparent to those with skill in the art, and the general principles defined herein may be applied to other embodiments. Therefore, the present invention is not intended to be limited to the particular embodiments shown and described, but is to be accorded the widest scope consistent with the principles and novel features herein disclosed.
(121) With respect to the use of substantially any plural and/or singular terms herein, those having skill in the art can translate from the plural to the singular and/or from the singular to the plural as is appropriate to the context and/or application. The various singular/plural permutations are not expressly set forth herein for sake of clarity.
(122) All of the methods described herein may include storing results of one or more steps of the method embodiments in memory. The results may include any of the results described herein and may be stored in any manner known in the art. The memory may include any memory described herein or any other suitable storage medium known in the art. After the results have been stored, the results can be accessed in the memory and used by any of the method or system embodiments described herein, formatted for display to a user, used by another software module, method, or system, and the like. Furthermore, the results may be stored “permanently,” “semi-permanently,” temporarily,” or for some period. For example, the memory may be random access memory (RAM), and the results may not necessarily persist indefinitely in the memory.
(123) It is noted herein that the one or more components of system may be communicatively coupled to the various other components of system in any manner known in the art. For example, the one or more processors may be communicatively coupled to each other and other components via a wireline connection or wireless connection. The wireless connection may include an unencrypted signal or an encrypted signal.
(124) The herein described subject matter sometimes illustrates different components contained within, or connected with, other components. It is to be understood that such depicted architectures are merely exemplary, and that in fact many other architectures can be implemented which achieve the same functionality. In a conceptual sense, any arrangement of components to achieve the same functionality is effectively “associated” such that the desired functionality is achieved. Hence, any two components herein combined to achieve a particular functionality can be seen as “associated with” each other such that the desired functionality is achieved, irrespective of architectures or intermedial components. Likewise, any two components so associated can also be viewed as being “connected,” or “coupled,” to each other to achieve the desired functionality, and any two components capable of being so associated can also be viewed as being “couplable,” to each other to achieve the desired functionality. Specific examples of couplable include but are not limited to

physically mateable and/or physically interacting components and/or wirelessly interactable and/or wirelessly interacting components and/or logically interacting and/or logically interactable components.

(125) Furthermore, it is to be understood that the invention is defined by the appended claims. It will be understood by those within the art that, in general, terms used herein, and especially in the appended claims (e.g., bodies of the appended claims) are generally intended as “open” terms (e.g., the term “including” should be interpreted as “including but not limited to,” the term “having” should be interpreted as “having at least,” the term “includes” should be interpreted as “includes but is not limited to,” and the like). It will be further understood by those within the art that if a specific number of an introduced claim recitation is intended, such an intent will be explicitly recited in the claim, and in the absence of such recitation no such intent is present. For example, as an aid to understanding, the following appended claims may contain usage of the introductory phrases “at least one” and “one or more” to introduce claim recitations. However, the use of such phrases should not be construed to imply that the introduction of a claim recitation by the indefinite articles “a” or “an” limits any particular claim containing such introduced claim recitation to inventions containing only one such recitation, even when the same claim includes the introductory phrases “one or more” or “at least one” and indefinite articles such as “a” or “an” (e.g., “a” and/or “an” should typically be interpreted to mean “at least one” or “one or more”); the same holds true for the use of definite articles used to introduce claim recitations. In addition, even if a specific number of an introduced claim recitation is explicitly recited, those skilled in the art will recognize that such recitation should typically be interpreted to mean at least the recited number (e.g., the bare recitation of “two recitations,” without other modifiers, typically means at least two recitations, or two or more recitations). Furthermore, in those instances where a convention analogous to “at least one of A, B, and C, and the like” is used, in general such a construction is intended in the sense one having skill in the art would understand the convention (e.g., “a system having at least one of A, B, and C” would include but not be limited to systems that have A alone, B alone, C alone, A and B together, A and C together, B and C together, and/or A, B, and C together, and the like). In those instances where a convention analogous to “at least one of A, B, or C, and the like” is used, in general such a construction is intended in the sense one having skill in the art would understand the convention (e.g., “a system having at least one of A, B, or C” would include but not be limited to systems that have A alone, B alone, C alone, A and B together, A and C together, B and C together, and/or A, B, and C together, and the like). It will be further understood by those within the art that virtually any disjunctive word and/or phrase presenting two or more alternative terms, whether in the description, claims, or drawings, should be understood to contemplate the possibilities of including one of the terms, either of the terms, or both terms. For example, the phrase “A or B” will be understood to include the possibilities of “A” or “B” or “A and B.”

(126) From the above description, it is clear that the inventive concepts disclosed herein are well adapted to carry out the objects and to attain the advantages mentioned herein as well as those inherent in the inventive concepts disclosed herein. While presently preferred embodiments of the inventive concepts disclosed herein have been described for purposes of this disclosure, it will be understood that numerous changes may be made which will readily suggest themselves to those skilled in the art and which are accomplished within the broad scope and coverage of the inventive concepts disclosed and claimed herein.

Claims

1. An active electronically scanned array comprising: a controller; a plurality of radio frequency integrated circuits, wherein the plurality of radio frequency integrated circuits are arranged in a lattice; a plurality of primary point-to-point serial interfaces; and a plurality of secondary point-to-

point serial interfaces, wherein each of the plurality of radio frequency integrated circuits include one primary input from the plurality of primary point-to-point serial interfaces and one secondary input from the plurality of secondary point-to-point serial interfaces, wherein each of the plurality of radio frequency integrated circuits include one primary output to the plurality of primary point-to-point serial interfaces and one secondary output to the plurality of secondary point-to-point serial interfaces; wherein the controller and the plurality of radio frequency integrated circuits are configured to transmit a plurality of network messages between the controller and the plurality of radio frequency integrated circuits over the plurality of primary point-to-point serial interfaces and the plurality of secondary point-to-point serial interfaces; wherein the plurality of network messages comprise: a read network message comprising a read command frame and a first plurality of data frames; a write network message comprising a write command frame and a second plurality of data frames; a write-all network message comprising a write-all command frame and a third plurality of data frames; and a sync network message comprising a sync command frame.
2. The active electronically scanned array of claim 1, wherein the read network message and the write network message use positional addressing to define an address of the first plurality of data frames and the second plurality of data frames.
3. The active electronically scanned array of claim 2, wherein the read command frame, the first plurality of data frames, the write command frame, the second plurality of data frames, the write-all command frame, and the third plurality of data frames each comprise a same frame size.
4. The active electronically scanned array of claim 3, wherein the first plurality of data frames immediately follow the read command frame, the second plurality of data frames immediately follow the write command frame, and the third plurality of data frames immediately follow the write-all command frame.
5. The active electronically scanned array of claim 4, wherein the plurality of radio frequency integrated circuits are configured to append the first plurality of data frames to the read network message in response to receiving the read command frame, wherein the plurality of radio frequency integrated circuits are configured to truncate the second plurality of data frames from the write network message in response to receiving the write command frame.
6. The active electronically scanned array of claim 4, wherein the read command frame, the first plurality of data frames, the write command frame, the second plurality of data frames, the write-all command frame, and the third plurality of data frames each comprise a start bit, a header, and a packet payload.
7. The active electronically scanned array of claim 6, wherein the packet payload of the read command frame, the write command frame, and the write-all command frame each comprise a plurality of address bits for the first plurality of data frames, the second plurality of data frames, and the third plurality of data frames, respectively.
8. The active electronically scanned array of claim 7, wherein the plurality of radio frequency integrated circuits each comprise: a first receiver, a first decoder, a second receiver, a second decoder, a router, a plurality of device registers, a first encoder, a first transmitter, a second encoder, and a second transmitter; wherein the plurality of address bits contain a register address of one or more of the plurality of device registers.
9. The active electronically scanned array of claim 8, wherein the plurality of address bits do not indicate to and from which of the plurality of radio frequency integrated circuits the plurality of network messages are addressed.
10. The active electronically scanned array of claim 8, wherein the controller is configured to read a health status message from the plurality of radio frequency integrated circuits by sending the read network message with the read command frame including the packet payload with the address at which a health status message is stored in the plurality of device registers.
11. The active electronically scanned array of claim 8, wherein the plurality of radio frequency integrated circuits are each configured to write the packet payload of the third plurality of data

frames into the plurality of device registers.

12. The active electronically scanned array of claim 8, wherein the first receiver and the second receiver receive serial input from the plurality of primary point-to-point serial interfaces and the plurality of secondary point-to-point serial interfaces, respectively.

13. The active electronically scanned array of claim 8, wherein the first decoder is coupled to the first receiver, wherein the second decoder is coupled to the second receiver, wherein the first decoder and the second decoder are each configured to decode a data signal to determine a command frame and one or more data frames of a network message.

14. The active electronically scanned array of claim 8, wherein the router is coupled to the first decoder, the second decoder, the plurality of device registers, the first decoder, and the second decoder.

15. The active electronically scanned array of claim 6, wherein the header comprises a plurality of bits indicating whether a frame is the read command frame, the write command frame, the write-all command frame, the sync command frame, or a data frame.

16. The active electronically scanned array of claim 1, wherein the plurality of primary point-to-point serial interfaces and the plurality of secondary point-to-point serial interfaces comprise a clock signal and a data signal, wherein one of: the clock signal and the data signal are on separate pins; or the clock signal is encoded onto the data signal using Manchester encoding.

17. The active electronically scanned array of claim 1, wherein the plurality of primary point-to-point serial interfaces are arranged orthogonal to plurality of secondary point-to-point serial interfaces.

18. The active electronically scanned array of claim 1, comprising a plurality of multi-chip modules; wherein the plurality of multi-chip modules comprise the plurality of radio frequency integrated circuits and a plurality of transmit/receive modules.

19. The active electronically scanned array of claim 1, wherein the plurality of radio frequency integrated circuits are arranged in a plurality of arrays, wherein the plurality of primary point-to-point serial interfaces and the plurality of secondary point-to-point serial interfaces connect the plurality of arrays.

20. An active electronically scanned array comprising: a controller; a plurality of multi-chip modules, wherein the plurality of multi-chip modules comprise: a plurality of radio frequency integrated circuits, wherein the plurality of radio frequency integrated circuits are arranged in a lattice; and a plurality of transmit/receive modules; a plurality of primary point-to-point serial interfaces; a plurality of secondary point-to-point serial interfaces, wherein each of the plurality of radio frequency integrated circuits include one primary input from the plurality of primary point-to-point serial interfaces and one secondary input from the plurality of secondary point-to-point serial interfaces, wherein each of the plurality of radio frequency integrated circuits include one primary output to the plurality of primary point-to-point serial interfaces and one secondary output to the plurality of secondary point-to-point serial interfaces; wherein the controller and the plurality of radio frequency integrated circuits are configured to transmit a plurality of network messages between the controller and the plurality of radio frequency integrated circuits over the plurality of primary point-to-point serial interfaces and the plurality of secondary point-to-point serial interfaces; wherein the plurality of network messages comprise: a read network message comprising a read command frame and a first plurality of data frames; a write network message comprising a write command frame and a second plurality of data frames, wherein the read network message and the write network message use positional addressing to define an address of the first plurality of data frames and the second plurality of data frames; a write-all network message comprising a write-all command frame and a third plurality of data frames; and a sync network message comprising a sync command frame.